



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin

Dear Editor,

I am a PhD student at Trinity College Dublin, and I am writing to submit our manuscript entitled “Bayesian Latent Class Regression and Variable Selection with Applications to Sleep Patterns Data” for consideration in the *Annals of Applied Statistics*.

Our paper addresses an important methodological gap in Bayesian latent class regression (LCR) by developing a framework that jointly performs predictor and item variable selection. We employ Pólya-Gamma data augmentation to enable efficient Gibbs sampling and extend this approach to facilitate variable selection for both regression coefficients and categorical items. Through simulation studies, we demonstrate accurate parameter recovery and successful identification of informative variables. We apply our methodology to paediatric sleep health data, identifying four clinically distinct sleep difficulty profiles and revealing the predictive role of Autism Spectrum Disorder diagnosis.

We believe this work is well-suited to *Annals of Applied Statistics* as it contributes both methodological innovation in Bayesian latent variable modelling and demonstrates meaningful practical applications in health research. The paper bridges statistical methodology and applied science, addressing real challenges in analysing categorical data with covariates.

This manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have approved the submission. We would be honoured to have our work considered for publication and welcome any feedback during the review process.

Yours sincerely,

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